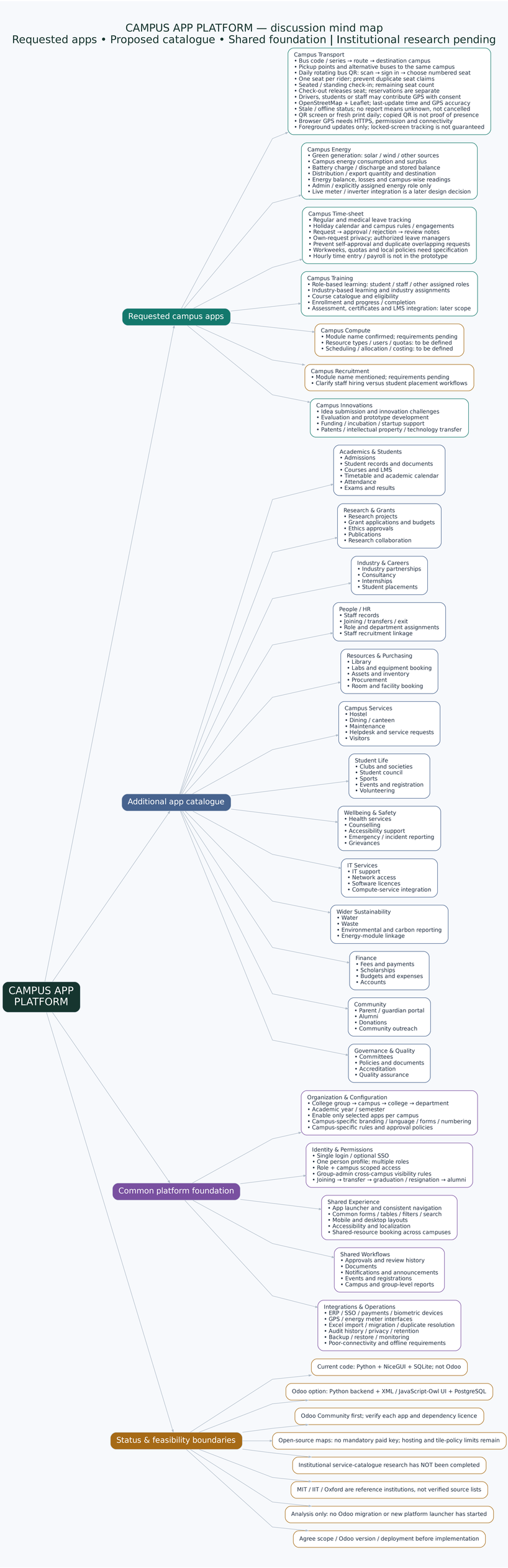
CAMPUS APP PLATFORM — discussion mind map
Requested apps • Proposed catalogue • Shared foundation | Institutional research pending
CAMPUS APP PLATFORM
Requested campus apps
Campus Transport
• Bus code / series → route → destination campus
• Pickup points and alternative buses to the same campus
• Daily rotating bus QR: scan → sign in → choose numbered seat
• One seat per rider; prevent duplicate seat claims
• Seated / standing check-in; remaining seat count
• Check-out releases seat; reservations are separate
• Drivers, students or staff may contribute GPS with consent
• OpenStreetMap + Leaflet; last-update time and GPS accuracy
• Stale / offline status; no report means unknown, not cancelled
• QR screen or fresh print daily; copied QR is not proof of presence
• Browser GPS needs HTTPS, permission and connectivity
• Foreground updates only; locked-screen tracking is not guaranteed
Campus Energy
• Green generation: solar / wind / other sources
• Campus energy consumption and surplus
• Battery charge / discharge and stored balance
• Distribution / export quantity and destination
• Energy balance, losses and campus-wise readings
• Admin / explicitly assigned energy role only
• Live meter / inverter integration is a later design decision
Campus Time-sheet
• Regular and medical leave tracking
• Holiday calendar and campus rules / engagements
• Request → approval / rejection → review notes
• Own-request privacy; authorized leave managers
• Prevent self-approval and duplicate overlapping requests
• Workweeks, quotas and local policies need specification
• Hourly time entry / payroll is not in the prototype
Campus Training
• Role-based learning: student / staff / other assigned roles
• Industry-based learning and industry assignments
• Course catalogue and eligibility
• Enrollment and progress / completion
• Assessment, certificates and LMS integration: later scope
Campus Compute
• Module name confirmed; requirements pending
• Resource types / users / quotas: to be defined
• Scheduling / allocation / costing: to be defined
Campus Recruitment
• Module name mentioned; requirements pending
• Clarify staff hiring versus student placement workflows
Campus Innovations
• Idea submission and innovation challenges
• Evaluation and prototype development
• Funding / incubation / startup support
• Patents / intellectual property / technology transfer
Additional app catalogue
Academics & Students
• Admissions
• Student records and documents
• Courses and LMS
• Timetable and academic calendar
• Attendance
• Exams and results
Research & Grants
• Research projects
• Grant applications and budgets
• Ethics approvals
• Publications
• Research collaboration
Industry & Careers
• Industry partnerships
• Consultancy
• Internships
• Student placements
People / HR
• Staff records
• Joining / transfers / exit
• Role and department assignments
• Staff recruitment linkage
Resources & Purchasing
• Library
• Labs and equipment booking
• Assets and inventory
• Procurement
• Room and facility booking
Campus Services
• Hostel
• Dining / canteen
• Maintenance
• Helpdesk and service requests
• Visitors
Student Life
• Clubs and societies
• Student council
• Sports
• Events and registration
• Volunteering
Wellbeing & Safety
• Health services
• Counselling
• Accessibility support
• Emergency / incident reporting
• Grievances
IT Services
• IT support
• Network access
• Software licences
• Compute-service integration
Wider Sustainability
• Water
• Waste
• Environmental and carbon reporting
• Energy-module linkage
Finance
• Fees and payments
• Scholarships
• Budgets and expenses
• Accounts
Community
• Parent / guardian portal
• Alumni
• Donations
• Community outreach
Governance & Quality
• Committees
• Policies and documents
• Accreditation
• Quality assurance
Common platform foundation
Organization & Configuration
• College group → campus → college → department
• Academic year / semester
• Enable only selected apps per campus
• Campus-specific branding / language / forms / numbering
• Campus-specific rules and approval policies
Identity & Permissions
• Single login / optional SSO
• One person profile; multiple roles
• Role + campus scoped access
• Group-admin cross-campus visibility rules
• Joining → transfer → graduation / resignation → alumni
Shared Experience
• App launcher and consistent navigation
• Common forms / tables / filters / search
• Mobile and desktop layouts
• Accessibility and localization
• Shared-resource booking across campuses
Shared Workflows
• Approvals and review history
• Documents
• Notifications and announcements
• Events and registrations
• Campus and group-level reports
Integrations & Operations
• ERP / SSO / payments / biometric devices
• GPS / energy meter interfaces
• Excel import / migration / duplicate resolution
• Audit history / privacy / retention
• Backup / restore / monitoring
• Poor-connectivity and offline requirements
Status & feasibility boundaries
Current code: Python + NiceGUI + SQLite; not Odoo
Odoo option: Python backend + XML / JavaScript-Owl UI + PostgreSQL
Odoo Community first; verify each app and dependency licence
Open-source maps: no mandatory paid key; hosting and tile-policy limits remain
Institutional service-catalogue research has NOT been completed
MIT / IIT / Oxford are reference institutions, not verified source lists
Analysis only: no Odoo migration or new platform launcher has started
Agree scope / Odoo version / deployment before implementation